'20.11.06.

Outine $\begin{cases} \text{partial trace, reduced density operator } (\sim \S 3.3) \\ \text{quantum operation } (\S 3.5) \end{cases}$

Thm 3.15 & Def 3.16 (partial trace)

Let $\mathbb{H}^A$, $\mathbb{H}^B$: Hilbert spaces with ONBs $\{|e_a\rangle\}$ and $\{|f_b\rangle\}$, and let $M \in L(\mathbb{H}^A \otimes \mathbb{H}^B)$ and $M_{a_1b_1,a_2b_2}$ be the matrix of M in ONB $\{|e_a \otimes f_b\rangle\}$.

Define $\begin{cases} \underline{tr^B(M)} := \sum_{a_1,a_2,b} M_{a_1b,a_2b} |e_{a_1}\rangle\langle e_{a_2}| \in L(\mathbb{H}^A) \\ \underline{tr^A(M)} := \sum_{b_1,b_2,a} M_{ab_1,ab_2} |f_{b_1}\rangle\langle f_{b_2}| \in L(\mathbb{H}^B). \end{cases}$

Then, we have

(i) $tr^B(M)$ and $tr^A(M)$ do not depend on the choice of ONBs.

(ii) $tr^B(M)$ and $tr^A(M)$ are unique in the sense that

$$\begin{cases} tr(M^A\, tr^B(M)) = tr((M^A \otimes \mathbb{1}^B) M) \quad \forall M^A \in L(\mathbb{H}^A) \\ tr(M^B\, tr^A(M)) = tr((\mathbb{1}^A \otimes M^B) M) \quad \forall M^B \in L(\mathbb{H}^B). \end{cases}$$

(3.47)

Here, tr^B is called as the partial trace over $\mathbb{H}^B$ and tr^A is called as the partial trace over $\mathbb{H}^A$.

(sketch of the proof)

(i) (independence on ONB). Let $\{|\tilde{e}_a\rangle\}$ and $\{|\tilde{f}_b\rangle\}$ be two other ONBs. Then, $\exists\, U^A \in \mathcal{U}(\mathbb{H}^A),\ U^B \in \mathcal{U}(\mathbb{H}^B)$

s.t. $|\tilde{e}_a\rangle = U^A|e_a\rangle$ and $|\tilde{f}_b\rangle = U^B|f_b\rangle$.

(ii) Verification of (3.47) : $M \in L(\mathbb{H}^A \otimes \mathbb{H}^B)$

$$M = \sum_{a_1 a_2 b_1 b_2} M_{a_1 b_1, a_2 b_2} |e_{a_1} \otimes f_{b_1}\rangle\langle e_{a_2} \otimes f_{b_2}| .$$

For $M^A \in L(\mathbb{H}^A)$, $M^A = \sum_{a_1, a_2} M^A_{a_1 a_2} |e_{a_1}\rangle\langle e_{a_2}|$

(iii) (Uniqueness) Let $\widetilde{tr^B(M)}$ be another operator on $\mathbb{H}^A$ satisfying $(3.47)_1$.

Then, $tr\left(M^A \left(\widetilde{tr^B(M)} - tr^B(M)\right)\right) = 0$

$=0$

Remark (i) $tr(\underbrace{M^A tr^B(M)}_{\in L(\mathbb{H}^A)}) = tr(\underbrace{(M^A \otimes \mathbb{1}^B)M}_{\in L(\mathbb{H}^A \otimes \mathbb{H}^B)})$

Thus, $tr^B(M)$ is a unique operator satisfying the relation above.

(ii) $tr : L(\mathbb{H}) \to \mathbb{C}$: complex number. : "trace"

$tr^B : L(\mathbb{H}^A \otimes \mathbb{H}^B) \to L(\mathbb{H}^A)$: "partial trace"

Exercise 3.42 For any $M \in \mathcal{L}(\mathbb{H}^A \otimes \mathbb{H}^B)$,

$$\mathrm{tr}(\mathrm{tr}^B(M)) = \mathrm{tr}(M) = \mathrm{tr}(\mathrm{tr}^A(M)).$$

Thm 3.17 & Def 3.18 (Reduced density operator).

Let $\rho \in D(\mathbb{H}^A \otimes \mathbb{H}^B)$ be a density operator on $\mathbb{H}^A \otimes \mathbb{H}^B$.

(i) Then, $\rho^A(\rho) := \mathrm{tr}^B(\rho)$ is the uniquely determined density operator.

(ii) For any bounded observable $M^A \in B_{sa}(\mathbb{H}^A)$,
(set of bounded and self-adjoint operator on $\mathbb{H}^A$)

$$\langle M^A \rangle_{\rho^A(\rho)} = \langle M^A \otimes \mathbb{1}^B \rangle_\rho.$$

(iii) $\{|e_a\rangle\}$, $\{|f_b\rangle\}$: ONBs in $\mathbb{H}^A$ and $\mathbb{H}^B$.
$\{|e_a \otimes f_b\rangle\}$ in $\mathbb{H}^A \otimes \mathbb{H}^B$.

$\rho_{a_1b_1,a_2b_2}$: matrix of ρ in ONB

Then, $$\rho^A(\rho)_{a_1a_2} = \sum_b \rho_{a_1b,a_2b} \quad \text{--- (3.52)}$$

Here, $\rho^A(\rho)$ is called as the reduced density operator.

(sketch of the proof)

(i) ($\rho^A(\rho)$: density operator)

- self-adj. : $(\rho^A(\rho)_{a_1a_2})^* = \rho^A(\rho)_{a_1a_2}$

$$(\because (\rho^A(\rho)_{a_1a_2})^* = \overline{\rho^A(\rho)_{a_2a_1}} = \overline{\sum_b \rho_{a_2b,a_1b}}$$

$$\overset{(\rho=\rho^*)}{=} \sum_b \rho_{a_1b,a_2b} = \rho^A(\rho)_{a_1a_2})$$

- positivity : For any $|\varphi\rangle \in \mathbb{H}^A$,

$$\langle \varphi | \rho^A(\rho) \varphi \rangle = \sum_{a_1, a_2} \overline{\varphi_{a_1}} \, \rho^A(\rho)_{a_1 a_2} \varphi_{a_2}$$

$$= \sum_{a_1 a_2 b} \overline{\varphi_{a_1}} \, \rho_{a_1 b, a_2 b} \varphi_{a_2} = \sum_b \langle \varphi \otimes f_b | \rho (\varphi \otimes f_b) \rangle \geq 0.$$

- $\mathrm{tr}(\rho^A) = 1$; $\mathrm{tr}(\rho^A(\rho)) = \mathrm{tr}(\mathrm{tr}^B(\rho))$

$$= \mathrm{tr}(\rho) = 1.$$

(ii) $\langle M^A \otimes \mathbb{1}^B \rangle_\rho = \mathrm{tr}((M^A \otimes \mathbb{1}^B)\rho) = \mathrm{tr}(M^A \, \mathrm{tr}^B(\rho))$

$$= \mathrm{tr}(M^A \rho^A(\rho)) = \langle M^A \rangle_{\rho^A(\rho)}$$

($\because$ The quantum mechanical expectation value of an observable A in a mixed state ρ $\langle A \rangle_\rho := \mathrm{tr}(\rho A)$.) ∎

Remark $\rho^A(\rho)$ is the state obtained from the state $\rho \in D(\mathbb{H}^A \otimes \mathbb{H}^B)$ by taking partial trace over subsystem B. ; $\rho^A(\rho) = \mathrm{tr}^B(\rho)$

Exercise 3.43. $M^A \in L(\mathbb{H}^A)$, $M^B \in L(\mathbb{H}^B)$.

$M^A \otimes M^B \in L(\mathbb{H}^A \otimes \mathbb{H}^B)$

(i) $\mathrm{tr}(M^A \otimes M^B) = \mathrm{tr}(M^A) \cdot \mathrm{tr}(M^B)$

(ii) $\mathrm{tr}^B(M^A \otimes M^B) = \mathrm{tr}(M^B) \, M^A$

(iii) $\mathrm{tr}^A(M^A \otimes M^B) = \mathrm{tr}(M^A) \, M^B$.

<u>Exercise 3.45</u> $\mathbb{V}$: finite dim'l v-sp. over a field $\mathbb{F}$.

$$L(\mathbb{V}^{\otimes n}) = L(\mathbb{V})^{\otimes n}$$

$\swarrow$

${}^{q}\mathbb{H}$: qubit Hilbert space $\simeq \mathbb{C}^2$.

a system of n qubits is described ${}^{q}\mathbb{H}^{\otimes n}$.

In general, any observable of such system or its operactor on n qubits.

$\Longleftrightarrow$ any such operator can be pepresented as a linear conbination of n-fold tensor products of operators on ${}^{q}\mathbb{H}$.

<u>Remark</u> This does not mean that every $A \in L(\mathbb{V}^n)$ is of the form

$$A = A_1 \otimes \cdots \otimes A_n \quad \text{for some } \{A_i\} \in L(\mathbb{V}).$$

$$A = \sum_{j} a_{j_1 \cdots j_n} A_{j_1} \otimes \cdots \otimes A_{j_n} . \quad \{A_{j_i}\} \in L(\mathbb{V}).$$

(sketch of the proof).

<u>claim</u> (i) $L(\mathbb{V}^{\otimes n}) \subseteq L(\mathbb{V})^{\otimes n}$ (ii) $\dim(L(\mathbb{V}^{\otimes n})) = \dim(L(\mathbb{V})^{\otimes n})$

(i) Any $w \in \mathbb{V}^{\otimes n}$, $w = \sum w_{j_1 \cdots j_n} v_{j_1} \otimes \cdots v_{j_n}$

$\{v_j\}_{j=1}^{\dim \mathbb{V}}$: basis of $\mathbb{V}$.

For any $A_1, \cdots, A_n \in L(\mathbb{V})$,

$$(A_1 \otimes \cdots \otimes A_n) w = \sum_{j_1 \cdots j_n} w_{j_1 \cdots j_n} (A_1 v_{j_1}) \otimes \cdots \otimes (A_n v_{j_n}).$$

$\Rightarrow \quad A_1 \otimes \cdots \otimes A_n \in L(V^{\otimes n})$

(ii) $\dim L(V) = (\dim V)^2$, $\dim (V^{\otimes n}) = (\dim V)^n$

$$\Rightarrow \dim (L(V)^{\otimes n}) = (\dim L(V))^n = (\dim V)^{2n}$$

$$\dim L(V^{\otimes n}) = (\dim V^{\otimes n})^2 = (\dim V)^{2n}. \quad \blacksquare$$

§ 3.5 Quantum operation

There are two ways that a quantum system can change

(i) by unitary time evolution by some Hamiltonian

(ii) a state transformation by a measurement

Motivation

1. For a system A of our interest, $\rho^A \in D(\mathbb{H}^A)$.

2. Combine (A, ρ^A) with the environment (B, ρ^B).

$$\iota_{\rho_B} : D(\mathbb{H}^A) \longrightarrow D(\mathbb{H}^A \otimes \mathbb{H}^B)$$
$$\rho^A \longmapsto \rho^A \otimes \rho^B$$

3. $U \in \mathcal{U}(\mathbb{H}^A \otimes \mathbb{H}^B)$; $U : D(\mathbb{H}^A \otimes \mathbb{H}^B) \longrightarrow D(\mathbb{H}^A \otimes \mathbb{H}^B)$

$$\rho^A \otimes \rho^B \longmapsto U(\rho^A \otimes \rho^B)U^*$$

4. P^B : projection onto the eigenspace of an observable of B.

$$\blacksquare = U(\rho^A \otimes \rho^B)U^* \longmapsto \frac{(\mathbb{1}^A \otimes P^B)\, \blacksquare\, (\mathbb{1}^A \otimes P^B)}{\mathrm{tr}((\mathbb{1}^A \otimes P^B)\, \blacksquare)}.$$

Note This measurement

$$\underset{\text{state before measurement}}{\rho} \longmapsto \frac{P \rho P}{tr(P\rho)} \quad : \text{ state after measurement.}$$

5. Finally, ignoring B, we can obtain a description of syst. A by taking the partial trace over B.

$$\frac{(\mathbb{1}^A \otimes P^B)\, \blacksquare\, (\mathbb{1}^A \otimes P^B)}{tr((\mathbb{1}^A \otimes P^B)\,\blacksquare)} \longmapsto \underbrace{\frac{tr^B\big(\boxed{(\mathbb{1}^A \otimes P^B)}\,\blacksquare\,(\mathbb{1}^A \otimes P^B)\big)}{tr((\mathbb{1}^A \otimes P^B)\,\blacksquare)}}_{(*)} \in D(\mathbb{H}^A).$$

$\in D(\mathbb{H}^A \otimes \mathbb{H}^B)$

ρ^A

Our goal is to represent $(*)$ in terms of suitable operators on $\mathbb{H}^A$ only.

Lemma 3.23 $\mathbb{H}$: Hilbert, $\{K_\ell\}_{\ell=1}^m \subseteq \mathcal{L}(\mathbb{H})$. Then, for any $\rho \in D(\mathbb{H})$,

the operator $K(\rho) := \sum_{\ell=1}^m K_\ell \rho K_\ell^*$ satisfies

(i) $K(\rho)^* = K(\rho)$: self-adj.

(ii) $K(\rho) \geq 0$; positive.

(iii) Moreover, for $\delta \in [0,1]$ and $\rho \in D(\mathbb{H})$.

$$\sum_{\ell=1}^m K_\ell^* K_\ell \leq \delta \mathbb{1} \iff tr(K(\rho)) \leq \delta$$

Thm 3.24 & Def 3.26 (Quantum operation).

$\mathbb{H}^A$: finite dim'l Hilbert, $\mathcal{K}: \mathcal{D}(\mathbb{H}^A) \to \underbrace{\mathcal{D}_{\leq}(\mathbb{H}^A)}$

$\qquad = \{\rho \in \mathbb{B}(\mathbb{H}^A) : \rho^* = \rho,\ \rho \geq 0,\ tr(\rho) \leq 1\}$.

Then, for $\delta \in (0,1]$, TFAE

(i) $\exists\ \mathbb{H}^B$: finite dim'l, $V \in \mathbb{B}(\mathbb{H}^A \otimes \mathbb{H}^B)$, $\rho^B \in \mathcal{D}(\mathbb{H}^B)$

s.t. $V^* V \leq \delta \mathbb{1}^{AB}$ and $\mathcal{K}(\rho^A) = tr^B(V(\rho^A \otimes \rho^B)V^*)$

(3.74)

(ii) $\exists \{K_\ell\}_{\ell=1}^m \subseteq \mathcal{L}(\mathbb{H}^A)$ s.t.

$\sum_{\ell=1}^m K_\ell^* K_\ell \leq \delta \mathbb{1}^A$ and $\mathcal{K}(\rho^A) = \sum_{\ell=1}^m K_\ell \rho^A K_\ell^*$

(3.76)

Here, a quantum operation $\mathcal{K}$ is a convex-linear map that can be represented by (3.74) or (3.76).

Moreover, (i) if $\mathcal{K}(\rho) = \sum_{\ell=1}^m K_\ell \rho K_\ell^*$ where $\{K_\ell\} \subseteq \mathcal{L}(\mathbb{H})$ with $\sum_{\ell=1}^m K_\ell^* K_\ell \leq \mathbb{1}$, then $\{K_\ell\}$: Kraus operators.

NOT unique

is called as the operator-sum representation.

(ii) $\mathcal{K}(\rho) = tr^B(V(\rho \otimes \rho^B)V^*)$ with $V^* V \leq \mathbb{1}$

↳ is called as the environment representation

Remark (i) Thm 3.24 does not assert that every convex-linear map does not have one of the equivalent forms (3.74) or (3.76).

(ii) $\boxed{\text{environment rep.}} \longleftrightarrow \boxed{\text{operator-sum rep.}}$

- concrete and easy to related real-word
- Mathematically inconvenient

- rather abstract
- useful for calculations.

Cor 3.27 $K : D(H) \to D_{\leq}(H)$: quantum operation with Kraus operators $\{K_l\}_{l=1}^{m}$.

For $\tilde{m} \geq m$ and $U \in \mathcal{U}(\tilde{m})$: unitary matrix.

$$\tilde{K}_j := \sum_{l=1}^{m} U_{jl} K_l \quad ; \quad \{\tilde{K}_j\}_{l=1}^{\tilde{m}} \ : \text{Kraus operators.}$$

Summary

1. initially $\rho \in D(H)$
2. Combine with another system (environment, error)
3. subject to $\begin{cases} \text{unitary time evolution} \\ \text{possible measurement.} \end{cases}$
4. $$\begin{aligned} D(H) &\longrightarrow D(H) \\ \rho &\longmapsto \frac{K(\rho)}{\operatorname{tr}(K(\rho))} \end{aligned}$$: with the help of a quantum operation K.

Reference [Quantum computation and quantum information, Nielsen and Chuang] ch8. Quantum-noise and quantum operations.